

Neural Bellman Operators

Response to the 4 October 2026 Referee Report

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R11 revision

We thank the referee for the careful review of R10 and the independent reconstruction of its continuous-capital bound. The report identifies the central distinction that motivates this revision: a guarantee supplied by a feasible schedule and an architectural tube does not measure what the fitted policy accomplishes. We address that distinction with a new policy-specific verification result and a new final computational design in the same capital economy. The title, broader economic subject, and original applications are retained. Their full equations and prior evidence are not deleted; the main article and supplement have been reorganized with an exact-source relocation map.

The report considered the R10 evidence commit beginning 8ded2629; the present revision starts from its review commit beginning 3ad498fa. The report is an owner-commissioned advisory review, not an editorial decision of the Econometric Society. The response below concerns its substantive mathematical and numerical objections.

Principal changes

Theorem 9 replaces the common tube allowance with an independently verified lower bound on a fitted policy's improvement relative to the analytical schedule. It combines an exact paired payoff identity, the explicit continuous-time transfer estimate in Proposition 8, a proved clipping-tail account, and a finite-family empirical Bernstein bound. The policy-specific endpoint depends on the saved policy and its fresh outcomes. It can distinguish a useful policy from an unsuccessful one inside exactly the same tube.

The deployment rule is explicit. A controller updates an internal state from observed innovations and holds its action between dates. The economic state remains the original continuous diffusion. This is not an unannounced identification of an Euler policy with continuously observed-state feedback. The guarantee is conditional statistical inference at each declared initial state, not a deterministic state-uniform PDE certificate.

Section 8 implements NBO evaluation using current-policy rollout and pathwise costate targets, followed by a separately differentiated actor update. Ninety primary fits cover ten seeds, three dimensions, and three methods. The final design includes stronger direct-policy and affine alternatives, longer budgets, tuned projected critic greedification, checkpoint frontiers, radius and architecture tests, initial-state stresses, and deliberately poor policies. The protocol distinguishes development from the independent final noise bank.

The final raw-data replay contains 90 primary policies and 230 two-sided evaluation records. Among the 30 primary NBO policies, 30 have positive lower improvement endpoints; their lower endpoints range from 0.000716 to 0.001193. The largest NBO regret upper endpoint is 0.026672.

The archive reports 0 training failures separately from unmet economic accuracy targets. The main tables and the complete supplement report direct and affine comparisons without suppressing competitive baseline outcomes.

Blocking comments

B1: The best guarantee does not require the learned actor or critic. We agree about the logical content of the R10 guarantee and preserve it as the analytical benchmark. It is no longer the sole guarantee attached to a fitted policy. The new endpoint is $G_d - L_\phi$, where L_ϕ is an independently computed lower improvement bound for the policy actually deployed. In contrast to $G_d + D_{d,\epsilon}$, it changes with the weights and the observed paired return distribution. When $L_\phi > 0$, the guarantee improves upon the schedule’s bound itself. The proof does not assume training succeeds. The critic is used to generate a candidate through policy evaluation, while verification remains independent of its fitted residual. This separation is intentional, rather than evidence that a critic residual was certified when it was not.

B2: The actor has no robust gain over the schedule and is more expensive. The final experiment now reports the schedule as the primary economic comparator and uses ten seed-level policies per dimension. A mean Euler difference is not treated as a continuous-time gain: the sampling, clipping, forward-arithmetic, and time-discretization terms are all subtracted before the improvement statement is made. All seed rows, including unsuccessful targets, are retained. Training, verification, and deployed decision costs are reported separately. The schedule remains the zero-training-cost choice. The economic question is whether the verified additional gain justifies a computational charge, not whether learning can be made costless. The R10 negative comparisons in dimensions twenty and fifty remain in the supplement and are explicitly discussed in the new main text.

B3: Weakly tuned comparisons and unstable direct greedification. The primary design compares the NBO actor with both direct policy optimization using the same nonlinear actor and an affine state-dependent policy. It records equal simulator visits but does not claim equal total work: five critic and five actor updates per NBO batch differ from a direct gradient step through the simulator. Checkpoint cost–accuracy frontiers and 600-iteration direct-policy and affine runs expose this difference. Additional direct-HJB critics are fitted at 200, 600, and 1,200 steps; a declared validation search selects their budget and conservative improvement fraction. A projected direct-full policy is labelled a hybrid, not an unrestricted direct-full solution. These experiments do not assert that a common update count proves NBO superiority. They allow simpler and actor-free alternatives to remain competitive in the reported record.

B4: The certificate cannot distinguish good and poor policies. Theorem 9 is policy-specific after training. Its finite-family guarantee includes the candidate-selection and sensitivity rows declared in the protocol. The certificate accepts neither a sampled maximum as a uniform bound nor a measured state range as a tail cutoff. The clipping envelope follows from bounded action separation and a Gaussian concentration calculation. The complete bias bound refers to the original economic diffusion. The raw arrays, output clipping counts, source weights, lower and upper payoff endpoints, and intermediate constants are stored. Zero and deliberately saturated corrections provide explicit checks that a shared tube does not produce a shared post-training assessment.

B5: Too few seeds, conditional confidence intervals, and missing Euler error. The number of primary training seeds is increased from three to ten. Every fitted policy is assessed separately. Across-seed ranges and standard deviations are reported as algorithmic variability, while the simultaneous statistical endpoints are conditional on the fitted objects. No confidence interval for an average over three actors is presented as training robustness. Proposition 8 explicitly bounds the policy-value transfer rather than extrapolating from grid differences. The 256-, 512-, and 1,024-cell records remain available as separate refinement diagnostics. Development used seed 2027 and a disjoint noise bank; the final protocol states when it was fixed and makes no retrospective preregistration claim.

B6: A model-specific theorem within a broad economic paper. The capital primitives and full adapted comparison class are preserved. The new theorem advances the verification of that application; it is not claimed for recursive utility, stopping problems, or games without their additional hypotheses. The original preference, recursive-utility, temporal-self, strategic, and trace sections are retained in full in the supplement, with a main-text account of their connection to the NBO maps. This keeps the paper’s subject intact while distinguishing a complete theorem–algorithm–experiment chain from the different verified objects in other applications. In particular, no capital result is substituted for a missing continuous-transfer term in the preference model.

B7: Economic significance and the depth of the numerical contribution. The revised economic question concerns the value of state-dependent consumption relative to the common schedule in the same dense nonquadratic economy. Equation 40 decomposes that value into production, curvature cost, and terminal dispersion. The tables report utility differences and a clearly defined externally financed flow-consumption-equivalent normalization. Positive lower gains also quantify a permissible implementation charge in utility units. This is a quantitative conclusion in the stated stylized model, not an invented population calibration. The post-training guarantee is attached to the actual measured policy rather than an arbitrary member of a safe class. The final comparisons determine the size of the gain and which simpler alternatives achieve comparable outcomes; the paper does not declare a universal actor advantage.

Major comments

M1: Title and abstract. The title *Neural Bellman Operators* is retained because the paper continues to develop the original evaluation–improvement–verification construction and its economic applications. The abstract now states the policy-specific result and the precise sampled-innovation deployment rule. It no longer relies on inheriting an architectural bound as evidence of learned economic improvement. The untrained schedule and its stronger structural benchmark remain explicit.

M2: Safety, solution accuracy, and method superiority. These are separated throughout. Theorem 7 concerns an architectural envelope; Theorem 9 gives a conditional, policy-specific economic endpoint; the algorithm and baseline tables compare executed methods and costs. None of these statements is used as a substitute for another. Passing the reproducibility checks is not journal acceptance or a theorem of algorithmic superiority.

M3: Schedule and tube economic conversions. The schedule’s bound and flow-equivalent conversion are printed alongside the original tube allowance and the new post-training endpoints. The normalization holds other payoff terms and the reference allocation fixed and is labelled externally financed. It is not described as a self-financing resource counterfactual.

M4: Radius–performance–guarantee frontier. The saved 0.05, 0.10, and 0.15 radius comparisons report both paired payoffs and policy-specific bounds. Frozen-correction rescaling is distinguished from fresh fitting. Each row receives the same declared statistical family allocation and its own action range, transfer bound, and tail envelope. The additional width-16 and width-64 actors are actually retrained.

M5: Diagnose direct-full failure. The revision adds budget and conservative-improvement selection for both direct critic families, and it retains controlled derivative-perturbation and action-saturation diagnostics. These diagnostics identify sensitivity of greedification; they are not labelled independent estimates of the true continuous critic-gradient bias. A direct-full critic followed by a radius projection is an explicitly named conservative hybrid. This provides a repair experiment without treating the failure of an untuned full greedy policy as evidence that the NBO actor is universally better.

M6: Approximate policy iteration and conservative improvement. The discussion now directly cites conservative policy iteration, safe policy improvement, and approximate modified policy iteration. The use of separate evaluation and improvement maps and the identity that converts a verified improvement into a smaller regret bound are not claimed as new principles. The specific contribution is the operational continuous-economy transfer and policy-specific error account, paired with the executed rollout-costate NBO implementation. The empirical Bernstein bound is attributed to its source rather than restated as a new concentration theorem.

M7: The critic’s role and diagnostic relationships. The actor uses a learned marginal continuation value fitted against current-policy rollout costates. A held-out value/costate diagnostic is recorded for every primary NBO seed. These stochastic-target discrepancies and their relationship to final payoff are reported separately from the certificate. They are not described as exact derivative errors. Verification deliberately does not trust the critic; the policy has to earn its independent payoff bound.

M8: State-distribution stress. The final design includes positive and negative mean shifts, increased dispersion, a joint shift–dispersion case, and a fixed Student- $t(3)$ quantile profile with heavier cross-sectional tails. All exact initial vectors, saturation frequencies, and internal-state coverage frequencies are saved. These are controlled pointwise stress profiles, not an inferred guarantee under every random initial-state law. The original economic diffusion remains unbounded; no observed state range is promoted into an invariant economic domain.

M9: Conditional arithmetic. The supplement places the complete contract in one subsection: exact-real interpretation of stored coefficients, binary64 elementary arithmetic and dot-product bounds, polynomial transcendental kernels, interval scalar and spectral calculations, inward action projection, forward rounding, and the ideal-iid sampling assumption. The optional compiled kernels disable fast-math transformations. Unit tests and independent scalar quadrature stress the

implementation, but the paper continues to call the claim conditional rather than machine-formal. A saved numerical certificate is not a formal verification of Python, the CPU, or the pseudorandom generator.

M10: Main-paper length and historical material. The main article is reorganized around the controlled economy, the NBO maps, the verification accounts, the capital theorem, the new policy-specific result, and the economic comparison. Full historical application sections and proof blocks are relocated byte-for-byte. The prior roots and a relocation map preserve their exact source and labels. No other paper folder or earlier revision is edited. The supplement contains both the prior adverse comparisons and the new seed-level record.

Reproduction and scope

The source and evidence commits, executed environment, complete result arrays, fitted weights and checkpoints, test outputs, generated table hashes, and compiled PDFs are bound by the revision manifests. The source branch and the final evidence/referee branches are distinct. Final-run numerical failures and unmet economic targets are different fields. No failed accuracy target is rewritten as a successful computation, and no unsuccessful seed is replaced by a more favourable one.

The result remains conditional statistical verification of a precisely specified controller in the retained capital economy. General optimizer convergence, a universal ranking of neural and direct methods, a calibrated empirical welfare claim, and continuous-transfer guarantees for the other applications are not inferred from it. These distinctions define what the revised results establish; they do not alter the paper's topic or substitute a different research project for the NBO paper.